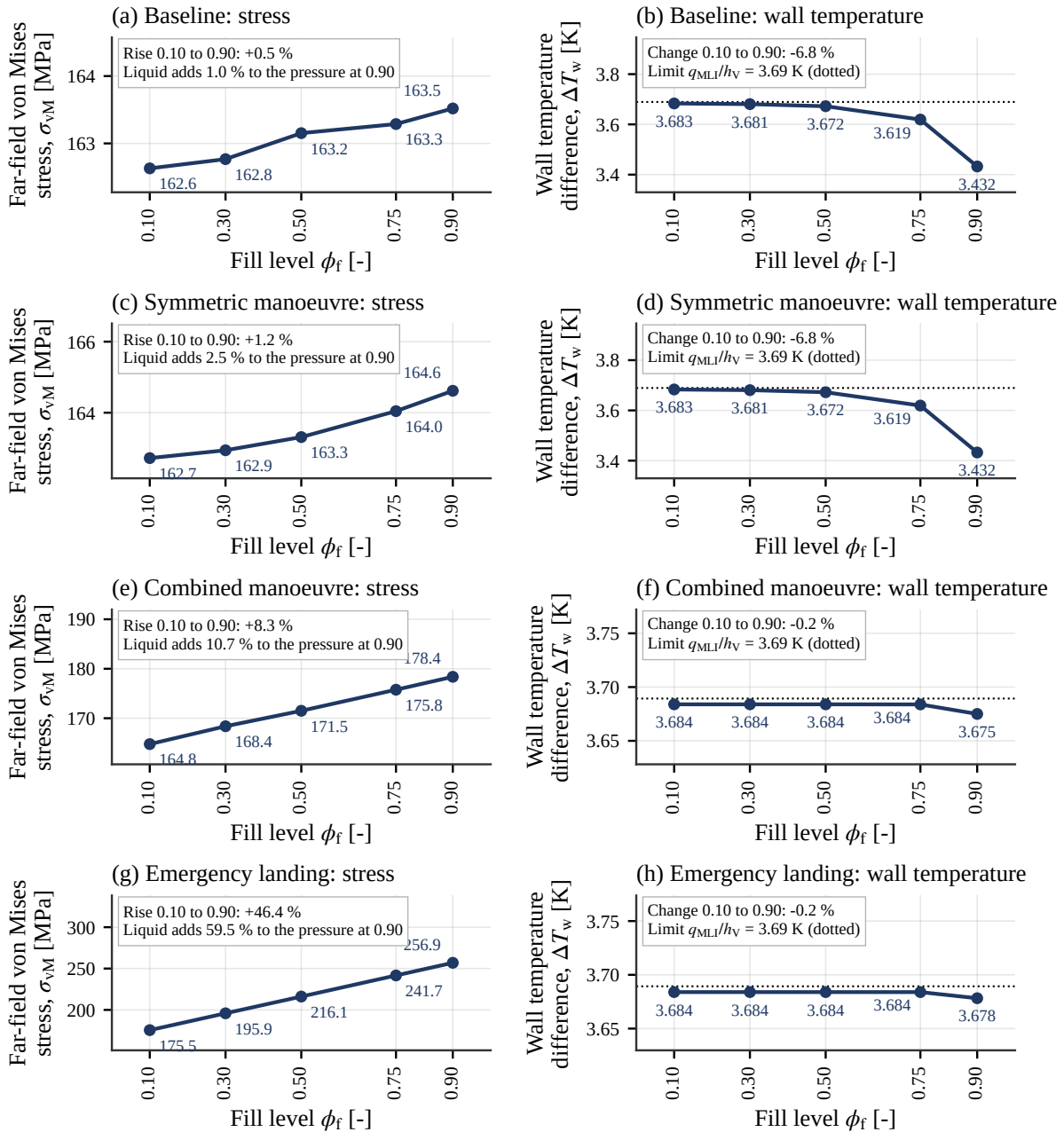
(a) Baseline: stress
Rise 0.10 to 0.90: +0.5 %
Liquid adds 1.0 % to the pressure at 0.90
Far-field von Mises stress, σ_{vM} [MPa]
164
163
162.6
162.8
163.2
163.3
163.5
0.10
0.30
0.50
0.75
0.90
Fill level ϕ_f [-]
(b) Baseline: wall temperature
Change 0.10 to 0.90: -6.8 %
Limit q_{MLI}/h_V = 3.69 K (dotted)
Wall temperature difference, ΔT_w [K]
3.8
3.6
3.4
3.683
3.681
3.672
3.619
3.432
Fill level ϕ_f [-]
(c) Symmetric manoeuvre: stress
Rise 0.10 to 0.90: +1.2 %
Liquid adds 2.5 % to the pressure at 0.90
166
164
162.7
162.9
163.3
164.0
164.6
Fill level ϕ_f [-]
(d) Symmetric manoeuvre: wall temperature
Change 0.10 to 0.90: -6.8 %
Limit q_{MLI}/h_V = 3.69 K (dotted)
3.683
3.681
3.672
3.619
3.432
Fill level ϕ_f [-]
(e) Combined manoeuvre: stress
Rise 0.10 to 0.90: +8.3 %
Liquid adds 10.7 % to the pressure at 0.90
190
180
170
164.8
168.4
171.5
175.8
178.4
Fill level ϕ_f [-]
(f) Combined manoeuvre: wall temperature
Change 0.10 to 0.90: -0.2 %
Limit q_{MLI}/h_V = 3.69 K (dotted)
3.75
3.70
3.65
3.684
3.684
3.684
3.684
3.675
Fill level ϕ_f [-]
(g) Emergency landing: stress
Rise 0.10 to 0.90: +46.4 %
Liquid adds 59.5 % to the pressure at 0.90
300
250
200
175.5
195.9
216.1
241.7
256.9
Fill level ϕ_f [-]
(h) Emergency landing: wall temperature
Change 0.10 to 0.90: -0.2 %
Limit q_{MLI}/h_V = 3.69 K (dotted)
3.75
3.70
3.65
3.684
3.684
3.684
3.684
3.678
Fill level ϕ_f [-]